Comixplain Helping Guide

COMIXPLAIN

This guide was created as part of the Comixplain research project, funded by the University of Applied Sciences St. Pölten (USTP) through the Innovation Call 2022.

Project Team:
Victor-Adriel De-Jesus-Oliveira
Hsiang-Yun Wu
Christina Stoiber
Magdalena Boucher
Alena Ertl

Contact:
victor.oliveira@fhstp.ac.at

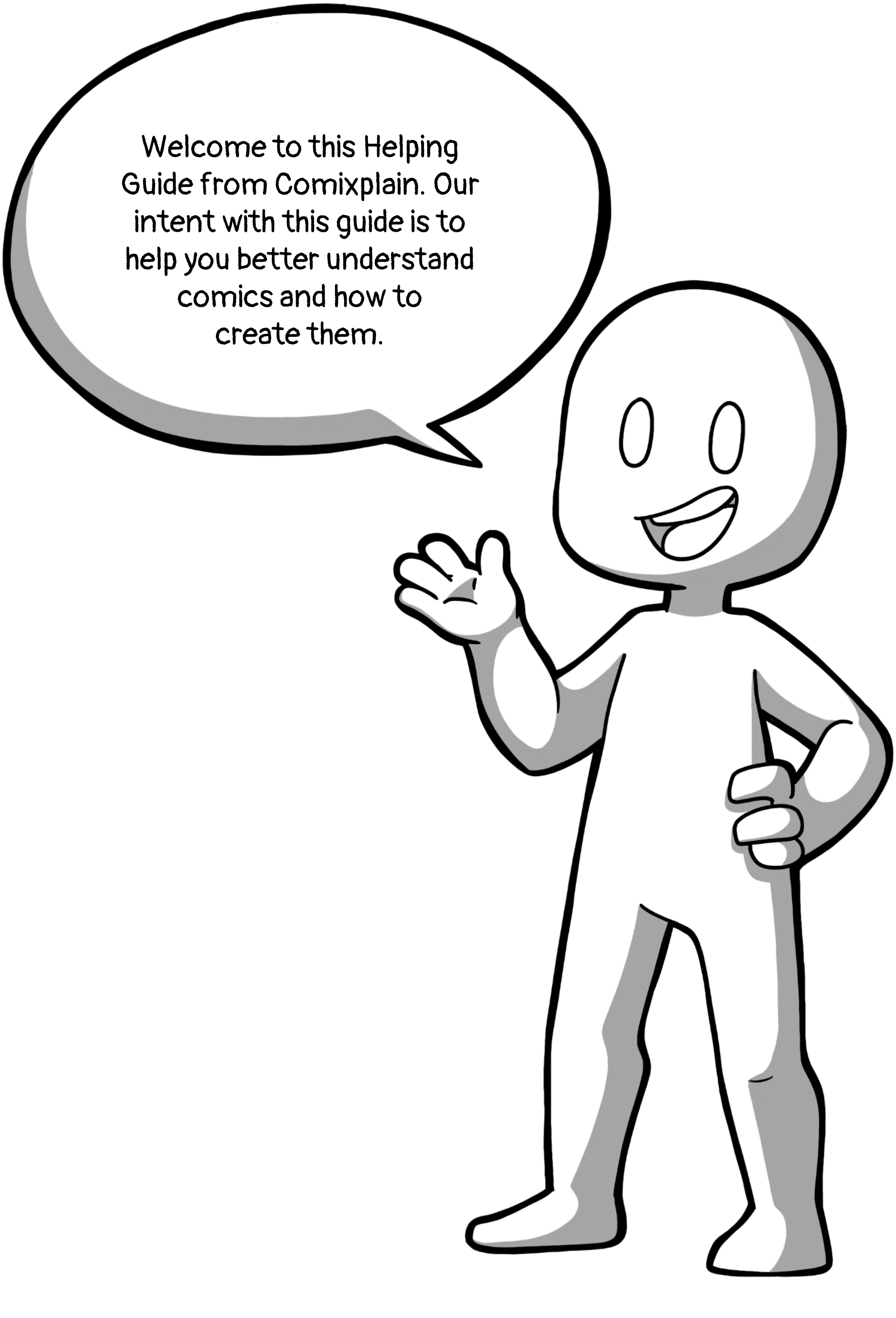
Authors:
Alena Boucher
Magdalena Boucher
Christina Stoiber
Hsiang-Yun Wu
Victor-Adriel De-Jesus-Oliveira

Illustrations:
Alena Boucher
Magdalena Boucher

1st Edition, 2023. Revised in June 2025.

University of Applied Sciences St. Pölten
EUDRES

https://www.**comixplain.cc**


Welcome to this Helping Guide from Comixplain. Our intent with this guide is to help you better understand comics and how to create them.

Content

Comixplain promotes student-centred teaching and learning. As students vary in their previous knowledge and preferences, we propose alternative didactic materials to cater to individual learning processes. We propose a novel storytelling-based strategy to engage students in learning complex scientific subjects through comics.

The effectiveness of comics in learning for both children and adults has been shown in many studies. However, many STEM topics have not yet been explored in depth. The power of comics in learning does not only come with their familiarity, but also from a unique combination of traits inherent to the medium itself: They offer an engaging narrative structure, similar to that found in learning videos, while maintaining the freedom of a 2D spatial layout, as seen in text, visualizations, and infographics. This means that a reader can benefit from a friendly, linear presentation and continuously keep the overview of the subject (as opposed to rewinding a video), and while having full control over the reading speed.

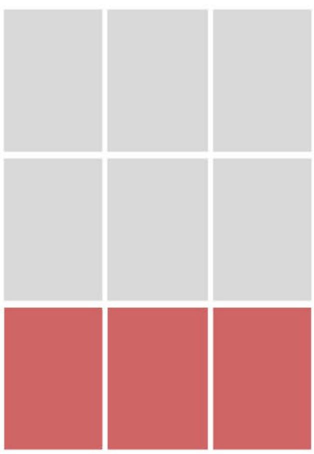
We hypothesize that comics can be used to effectively understand, review, or provoke the discussion of complex subjects such as math, statistics, or physics. The goal of this project was to understand and identify which factors are related to the difficulty in learning such topics and how comics can contribute to the learning process.

This guide will help you understand how to implement our comics in your own lectures and teaching activities. We explain our modular system and best practices to help you use and adapt our comics, as well as create new ones with the assets we provide in our repository (https://github.com/fhstp/comixplain).

Table of Contents

Panels are the main structure of comics. They are the borders around the drawings.

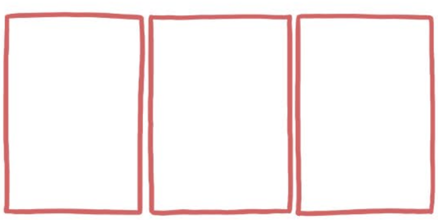
For Comixplain, we already have a base 3x3 grid that defines the shape and size of panels, but there are still a few design, storytelling, and general comic rules we should follow when placing them.



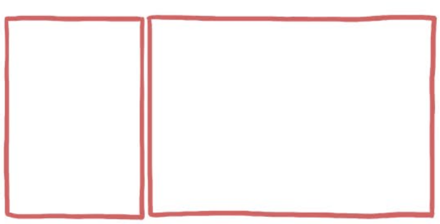
The grid provided with the Comixplain assets has three rows, each consisting of three squares. Each panel can be as small as one square, and as big as three in one row.

These rows in Comixplain are so-called **strips**.

Panel sizing in Comixplain according to the grid:



Three single panels with the same size

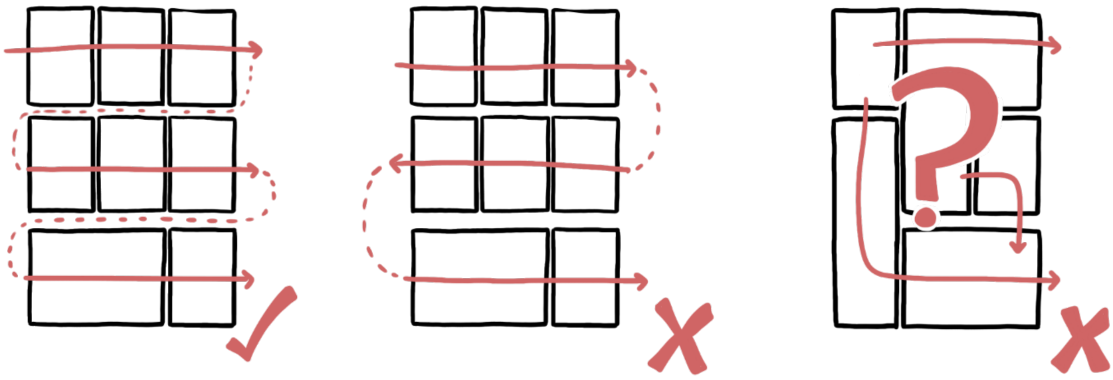


One single and a double-size panel, which spans over two singles. The order of those panels can be changed.

One triple-size panel for the full row.

Panels

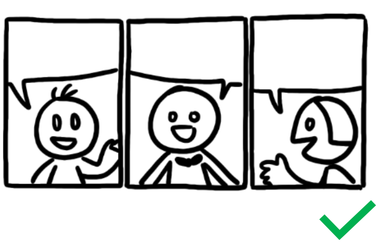
Western comics are read from left to right and top to bottom.
We follow this rule in comics too. Therefore, when reading a comic, we start with the top panel in the left corner, then read the first row from left to right, and continue reading like we would read a book.

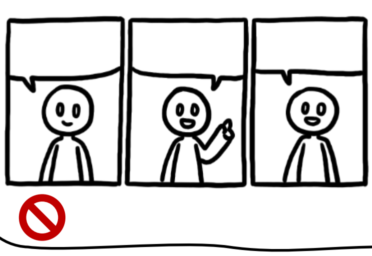


If we don't follow this rule by changing the reading direction or arranging the panels in a way that makes the reading order unclear, we confuse the readers and break their reading flow.

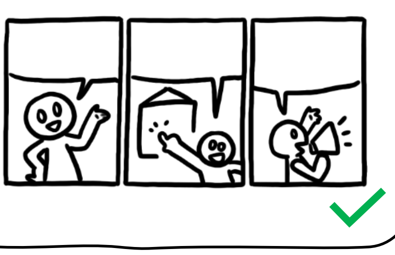
Another element we have to keep in mind is how to break the story down into individual panels.

Try to avoid overcrowded panels, and rather stretch the dialogue across multiple single panels. Then again, don't overdo it with too many single panels - try to achieve a pacing that feels natural to you.

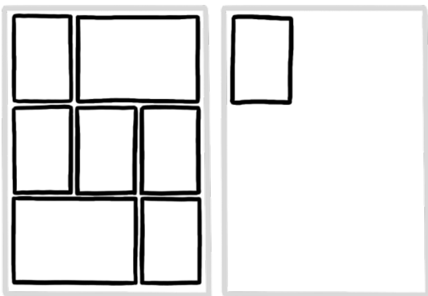




If you want to put more variation and life into the comic, try varying the poses of the characters.

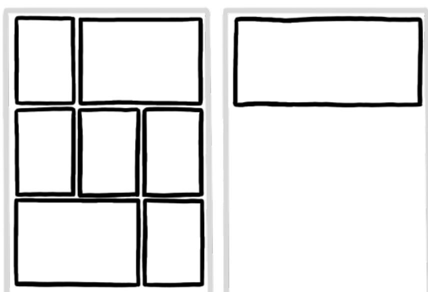


The panel arrangement should be applied logically over multiple pages.

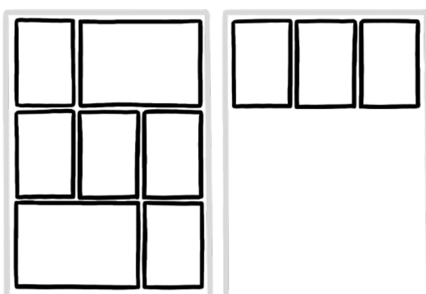


Try not to have just one small panel on a single page. The panel will seem out of place.

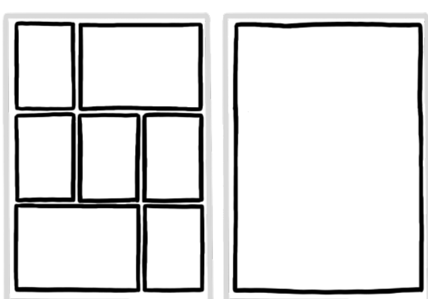
Do this instead:



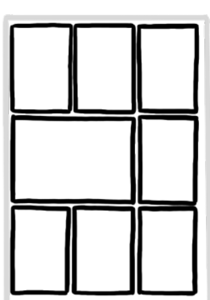
One full row panel



Lengthen the story to fill the row wth more single panels, or a single and a double-sized panel.



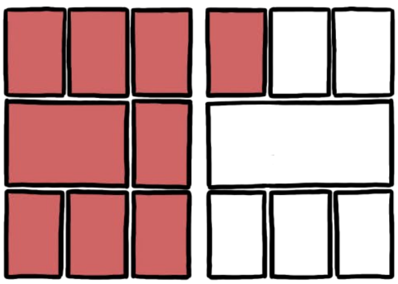
A full-page panel.
Though this makes only sense if you want to end the comic with an exercise and add an annotation space for the readers.

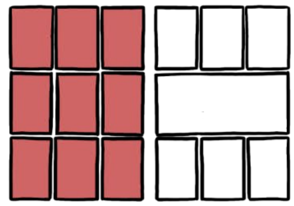


Rearrange panels and scripting to downsize your comic, and avoid using another page.

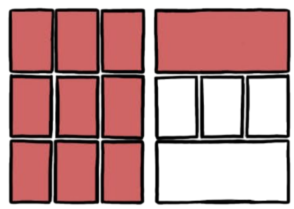
Just as we try not to have single panels on pages or to fill out rows, we want to keep separate **themes**, concepts, or topics in a comic together and apply shifts into other themes logically.

If one theme bleeds into another...





...make the change between the pages

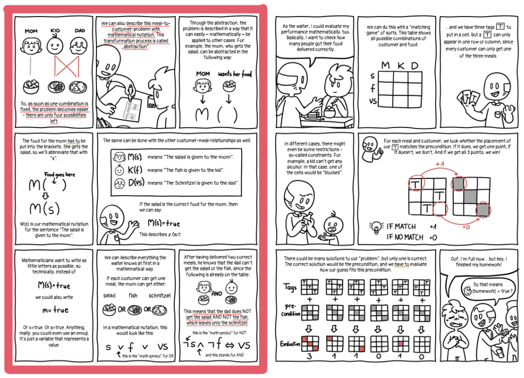


...use the full row of panels for the first topic (singles, doubles or whole row panel)

Example:

In the Comixplain comic "[Mathematical Thinking]", there are two different themes of the topic in the story: 1) Notation; and 2) Evaluation.

In the initial sketch, those two topics overlapped, and even if they are part of the same theme, they were two different components of the mathematical thinking explanation. Therefore, we rearranged the panels and text to better differentiate between the two.



For dialogues in comics, we use so-called **speech bubbles**.
There are many different types and styles of speech bubbles.
The most well-known examples are:

Normal Speech
These bubbles contain conversations or monologues that are spoken in a normal voice volume.

Thoughts
These contain thoughts of characters that are not spoken and are a tool to show the inner thoughts of characters to the reader.

Shouting
It includes any form of louder sounds. Can be happy, sorrowful, angry, surprised, and other versions of shouting. It just means that it is louder than normal.

For Comixplain, we mostly use variations of the normal speech bubble. In this helping guide, we will show you how to use the speech assets provided to you in PowerPoint, and some additional general rules for usage and layout of speech bubbles in comics.

These are the three used versions of normal speech bubbles in Comixplain:

Full Bubble

Half Bubbles (on top of panels)

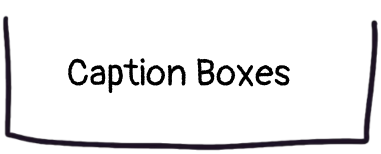
Boxes on top of panels for captions or narrations

These are the speech bubble assets we provide in the Comixplain catalog. Here, we will tell you how to use them in PowerPoint after importing them into your presentation.

First, always make sure that speech bubbles are under the panel layer, but **above the drawings layer** (see more in the layering chapter of this helping guide).

Full Bubble
You can place these easily on the comic.
They come with a white background so they can hide the artwork behind them, making lettering on them easy. Just make sure that the little triangle points to the person speaking the text in the speech bubble.

Half Bubble and Caption Boxes
The former are used for speech, while the boxes can have any written text, like narrations, thoughts, and mentions of time and space, on them. They are often used if a character is explaining or recounting something, and you see the talked-about topic instead of the character. Or if you want to explain the setting of the story, you can write the date or place in this box.

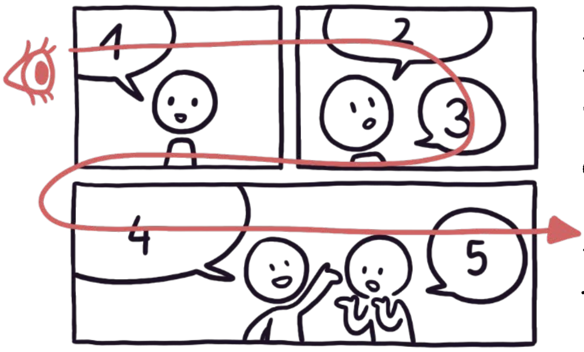


These two versions of asset bubbles have an open top border and are meant to be placed at the top of the panel. Because of the open top, you can move them freely from the top to the bottom and create a box that way. Because of this flexible use of the height of the boxes, only a small part of the Bubbles have a white background, so if you want to make them bigger, but there is artwork behind them that does not get hidden by the white background, you will need the help of creating white rectangles between the bubbles and the artwork.

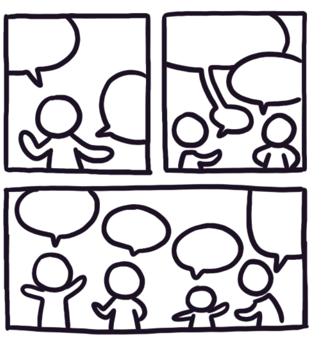
Speech Bubbles

Some basic speech bubble rules:

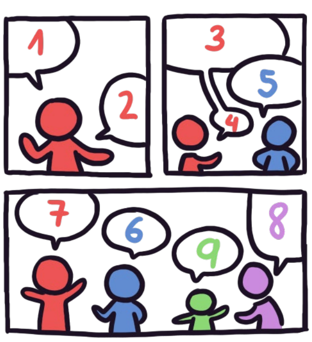
1) Reading Flow



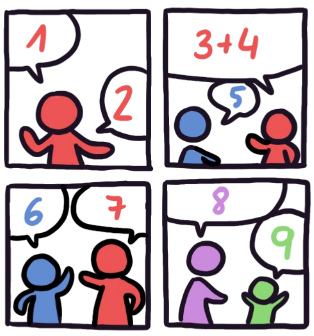
Just as with panels, we read speech bubbles from left to right, top to bottom, in the Western world. Therefore, we must arrange our speech bubbles following that rule. Reading starts in the top left corner, going from there towards the right, always reading the closest bubble.



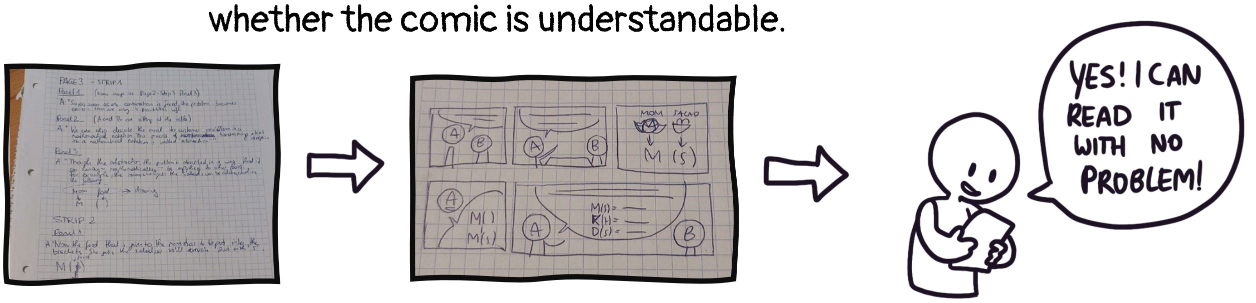
This comic would be read in the following way:



But if we want to read the bubbles how they are numbered in the colored version, we will need to rearrange the bubbles or the panel layout.

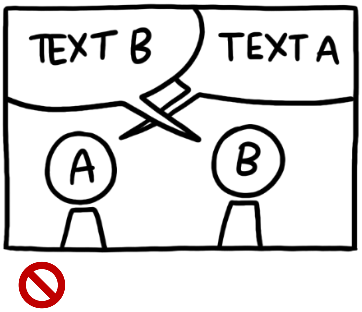


TIP Write a script of the comic first, then draw an initial and very basic sketch (can be crude), then show it to other people to get their opinion on whether the comic is understandable.



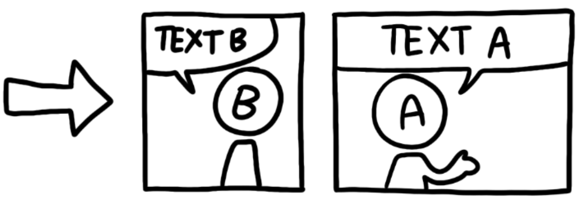
Speech Bubbles

2) Do not cross speech bubbles over each other.

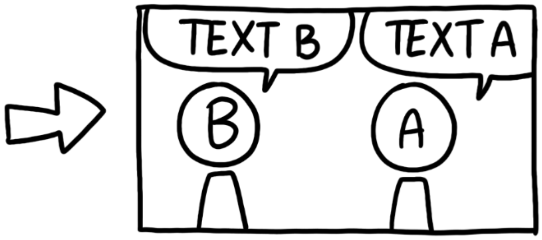


Since our reading direction is from left to right, we first read the bubble with "Text B". But, since our "artwork reading" is also from left to right, we associate the bubble with Person A instead of Person B. Because there are two speech bubbles, we tend to stick each of those to the closest character. Therefore, if the bubbles are crossed, it will throw off the readers.

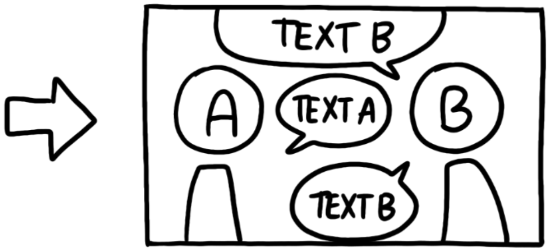
Instead, do one of the following versions:



Split the one panel into two. This way each bubble has it's own panel and the reading order is clear

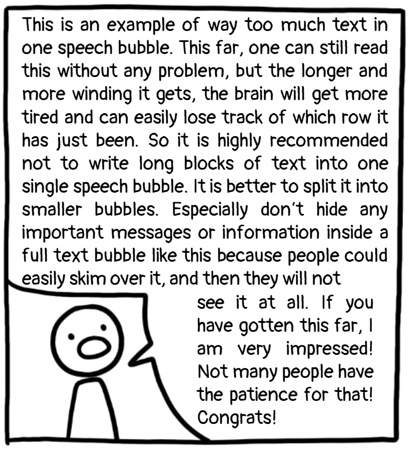


Change the position of A and B. Just be mindful of too many position jumps. If you do this too often, it could confuse the reader with how often the camera angle is moved.

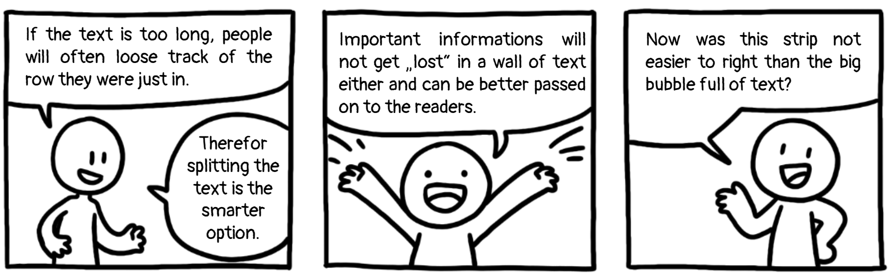


Change the reading direction from up to down. This only makes sense if you can change the script in a way where characters are talking in short sentences and alternating between each other. This version may also clutter the panel, so be mindful of its use.

3) Too much text in one speech bubble



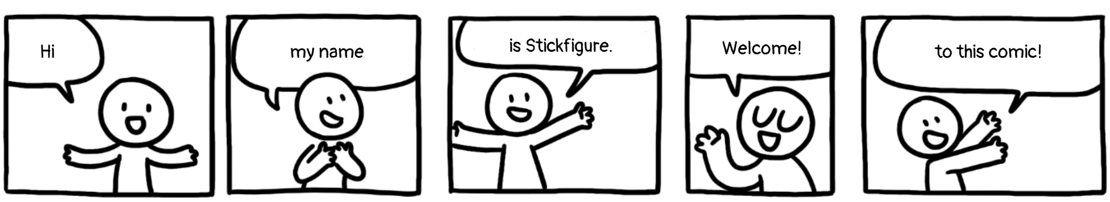
If there is too much text and information in one single speech bubble, people will often just skim over it and not read it properly or at all.
Imagine a really long text in a book that spans over two pages without any paragraphs or breaks after some sentences. Tiring to read, isn't it?

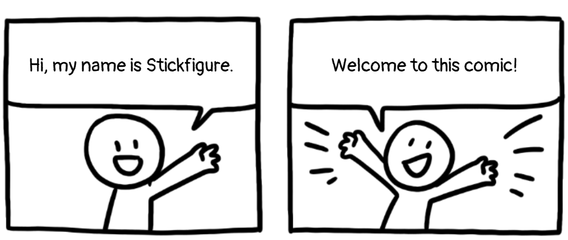


Better split that text into more panels!

But at the same time, do not overdo stretching your text!
You can lose your readers by dragging out your writing too.

If something can be said in just one or two panels without any reading problem, do not extend it unnecessarily just to get your comic longer.



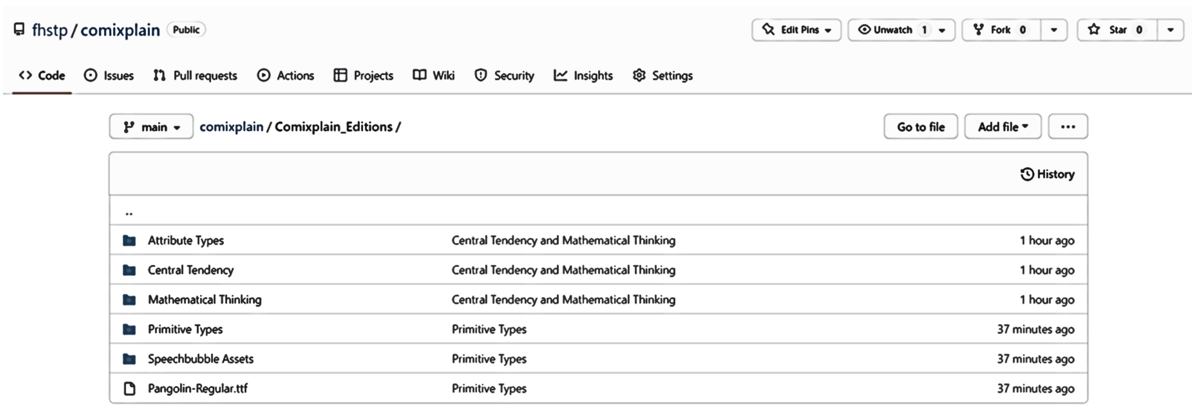


Layering

In this part of the guide, we will show you how to import our assets into PowerPoint and how to layer them.

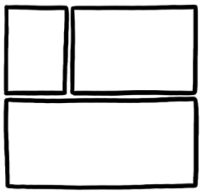
1) Import

In the Comixplain repository (https://github.com/fhstp/comixplain), you will find PowerPoint files with the basic 3x3 grid, and folders with assets, such as speech bubbles, characters, and comics font.

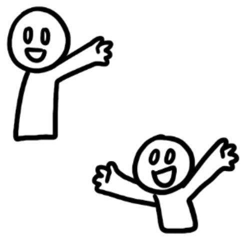


2) What do we need now for a comic?

Comixplain Grid (for reference of placement)



Borders for panels



People Assets

Background Assets

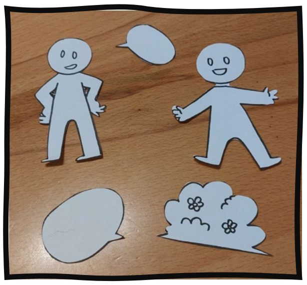
Speech bubbles

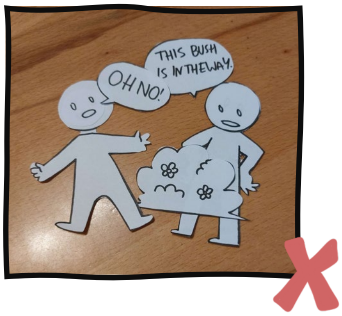
To put them all together we will need

Layering

But what is layering exactly?

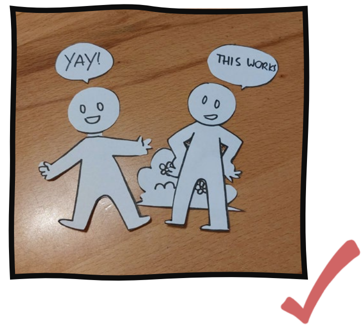
Just imagine you actually do the comic traditionally, and all the assets are cut out of paper. Now, when you lay them in a random order over each other, some parts of the assets will be hidden by other assets.



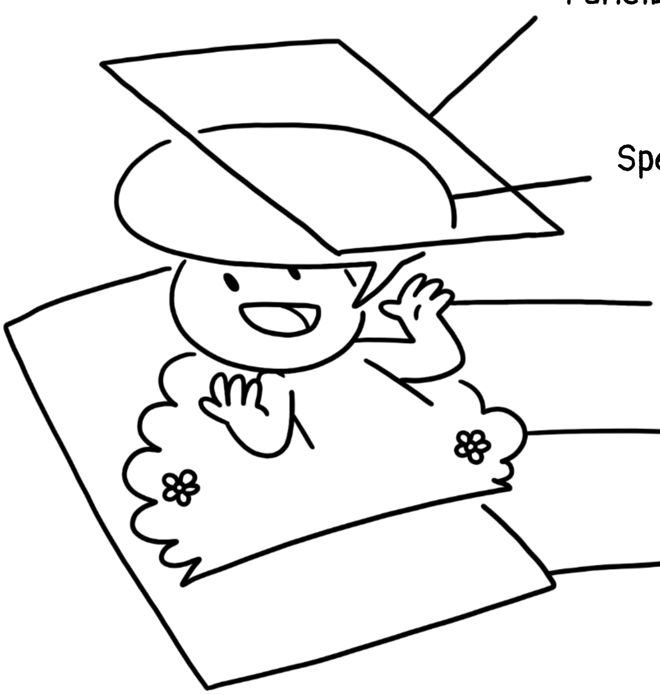


So we need to layer them in a order that makes sense

→



Layering order used in PowerPoint for Comixplain:



The text on the speech bubbles can be either placed between the panel and the speech bubble layer or on top of all the layers.

Finally, for **accessibility**, make all non-textual elements decorative only by using the Alt Text feature on PowerPoint.

How to cite this document:

Boucher, A., Boucher, M., Stoiber, C., Wu, H.-Y., & De-Jesus-Oliveira, V.-A. (2025). Comixplain helping guide (1st ed., rev. November 2025). University of Applied Sciences St. Pölten (USTP). https://www.comixplain.cc

This guidebook was adapted from: